

ElectroSim-LSV-RRDE-Analyzer

Publication-ready electrochemical analysis in the browser — upload a file, get the numbers and the figure.

Live app: <https://lsv-analysis-ir-compensation-tafel-slope.streamlit.app/>

A Streamlit application that takes raw potentiostat exports (Excel/CSV/ZIP) and returns the quantities electrocatalysis papers actually report — R_u , iR -corrected curves, onset potentials, overpotentials at benchmark current densities, Tafel slopes, Koutecký–Levich electron numbers, half-wave potentials, and RRDE peroxide yields — together with journal-styled plots and export-ready tables. No scripting, no installation, no data leaving memory.

Status

Aspect	Current state
Release	v1.1.0 — stable, actively maintained
Deployment	Public Streamlit Cloud app (link above); runs locally the same way
Scope	5 independent analysis modules, ~5.5 k lines of Python
Data handling	In-memory only; optional password gate; one-click session reset
License	MIT

Recent additions: batch ZIP loading of whole RRDE data folders, replicate-group statistics (mean $\pm$ SD) for LSV fits, reaction-aware automatic Tafel-region detection, stacked ring/disk multi-rotation-rate plots, and independent but linked EIS and LSV iR -correction tabs.

Capabilities

Module	Input	Output
EIS / R_u Analysis	Nyquist data (Z'' , Z''')	R_u from a high-frequency circle fit, plus R_{ct} and R_u+R_{ct} ; adjustable fit range or manual R_u
LSV iR Correction	Polarization curve + R_u	$E_{corr} = E - (f/100) \cdot I \cdot R_u$ at any factor 5–100 % (85 % recommended), with an over-compensation "fold-back" guard

LSV Analysis	One or many Potential, Current files	Onset potential, eta at benchmark j (e.g. 10 mA/cm ²), auto-detected Tafel region, Tafel slope + R ² + mechanistic benchmark, multi-sample overlays
K-L Analysis	RDE curves at 3+ rotation rates	Koutecký–Levich fit at several potentials -> kinetic current density j _k and electron-transfer number n
ORR / RRDE Analysis	Disk-only or disk + ring, 1+ samples, 1+ rotation rates	Onset, half-wave potential E _{1/2} , mass-transport-corrected Tafel slope, electron number n and %H ₂ O ₂ from ring/disk current, merged multi-rpm ring/disk figures

Cross-cutting features:

- **12 reactions supported** — HER, HOR, OER, ORR, CO₂RR, CORR, N₂RR, NO₃RR, MOR, EOR, UOR, or unspecified. Samples of different reactions can share one overlay, split into per-reaction legend groups; overpotentials are reported against the correct equilibrium potential where one is defined.
- **9 reference electrodes** — SHE/NHE, SCE, Ag/AgCl, Hg/HgO, Hg/Hg₂SO₄ variants — converted to RHE with $E(\text{RHE}) = E_{\text{meas}} + E^{\circ}(\text{ref vs NHE}) + 0.0592 \times \text{pH}$.
- **Real unit handling** — A <-> mA <-> μA <-> nA and their /cm² counterparts, auto-detected from column headers; the electrode area reconciles absolute current with current density and Ohm with Ohm·cm², so I·Ru always resolves to volts.
- **Replicate groups** — repeat scans sharing a group name get every fitted value reported as **mean ± SD** in its own results section.
- **Publication-styled export** — Arial fonts, selectable size, closed box axes, draggable legends and slope labels; download every plot as 300 dpi TIFF or interactive HTML, and every table as TIFF or CSV.

Quick start

```
pip install -r requirements.txt
streamlit run app.py
```

Open <http://localhost:8501>. Choose **Use bundled sample** in the sidebar to try the EIS/LSV tabs instantly, or load `sample-data/Example ORR 0-1 M KOH/` for a full five-rotation-rate RRDE dataset.

TIFF export needs headless Chrome (via kaleido); if it is missing, run `plotly_get_chrome` or use the HTML/CSV downloads.

Loading data

- **EIS / LSV tabs** — one Excel workbook (Sheet 1 = EIS Z' , Z'' ; Sheet 2 = LSV Potential, Current) or two CSVs. Several samples can sit side-by-side as repeated column pairs; pick the active one in the sidebar.
- **LSV Analysis** — one Potential, Current file per sample (Excel or CSV), combined into a single overlay.
- **K-L and ORR/RRDE** — share one loader, in two styles:
- **Per-sample uploaders** for raw per-electrode files exactly as instrument software exports them (Disk Current vs Disk Potential (1600 RPM).csv), or a compiled workbook with Potential, Disk current, Potential, Ring current already paired. Rotation rate and disk/ring role are guessed from the filename and can be corrected by hand.
- **Batch ZIP upload** — zip a data folder (one subfolder per sample) and load everything at once, roles and rotation rates read from filenames.

Electrolyte O_2 transport parameters for K-L ship as presets for 0.1 M KOH, 0.1 M $HClO_4$, and 0.5 M H_2SO_4 , or can be entered manually.

Methods

```
iR correction      E_corr = E - (f/100) · I · Ru
Koutecky-Levich    1/j = 1/j_k + 1/(B · w^0.5),  w = 2 · pi · rpm/60,
                  B = 0.62 n F D^(2/3) v^(-1/6) C
RRDE               n = 4|Id| / (|Id| + |Ir|/N)
                  %H2O2 = 200(|Ir|/N) / (|Id| + |Ir|/N)
```

- Koutecký, J.; Levich, V. G. *Zh. Fiz. Khim.* **1958**, 32, 1565.
- Bard, A. J.; Faulkner, L. R. *Electrochemical Methods*, 2nd ed.; Wiley, 2001; Ch. 9 (Levich constant B, RRDE collection efficiency, n and $\%H_2O_2$).

Full technical notes — data formats, security hardening, and using the Python API without the GUI — are in the module docstrings under [scripts/modules/](#).

Citation

If you use this app (including the public Streamlit deployment) in your work, please cite it. Machine-readable metadata is in [CITATION.cff](#).

Kumar, R. (2026). *ElectroSim-LSV-RRDE-Analyzer* (v1.1.0) [Computer software]. North Carolina Central University. DOI: 10.5281/zenodo.XXXXXXX.
<https://github.com/rajeev4187/LSV-Analysis-iR-compensation-Tafel-slope>

```
@software{kumar_electrosim_lsv_rrde_2026,  
  author = {Kumar, Rajeev},  
  title  = {ElectroSim-LSV-RRDE-Analyzer},  
  version = {1.1.0},  
  year   = {2026},  
  doi    = {10.5281/zenodo.XXXXXXX},  
  url    = {https://github.com/rajeev4187/LSV-Analysis-iR-compensation-Tafel-slope}  
}
```

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